3.5.1 ER Modeling basic concepts

The basic concepts of ER model consists of

1. Entity or Entity type
2. Attributes
3. Relationship

These are the general concepts which help to create an ER diagram and produce an ER model. With the help of these any database design can be created and viewed to know the concept in that database design.

3.5.2 Entity or Entity type

An Entity can be anything a real-world object or animation which is easily identifiable by anyone even by a common man.

Eg: In a company's database Employee, HR, Manager are considered as entities, where each of these entity will be having their own attributes. An entity is represented by a rectangular box. See Figure 3.5.



Figure 3.5 Entity

Types of Entity:

1. Strong Entity
2. Weak Entity
3. Entity Instance

Strong Entity

A Strong entity is the one which doesn't depend on any other entity on the schema or database and a strong entity will have a primary key with it (i.e. a unique id which other entities will not have in their attributes).It is represented by one rectangle.In the above example it is a strong entity because it has a primary key(a unique id) as the roll no because for every one roll no varies and it will not be same.

Weak Entity

A weak entity is dependent on other entities and it doesn't have any primary key like the Strong entity. It is represented by double rectangle.

For Example:

Here the marks is the weak entity and there are no unique id or primary key for that entity. So they are dependent on the existence of the other entity. See Table 3.3

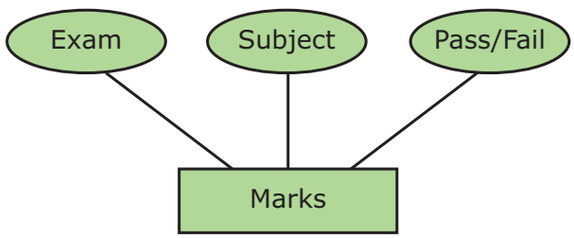


Figure: 3.6 Weak Entity

3.5.3 Entity Instance

Instances are the values for the entity if we consider animals as the entity their instances will be dog, cat, cow… Etc. So an Entity Instance denotes the category values for the given entity.

Table: 3.3 Entity Instances

Entity	Instances
Human	Male , Female
Animals	Dog, cats, lion
Jobs	Engineer, Doctor, Lawyer
Actors	Ajith, Vijay, Vikram
Electronics	Laptop, Mobile

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